
WORK EXPERIENCE

Applied LLM Engineer – GenAI Systems & Evaluation, Balanx-Bio | Jan 2026 – Present

- Built Pinecone and LoRA RAG pipelines using HuggingFace, reducing hallucinations **38%** while converting complex healthcare data into validated structured outputs supporting downstream analytics under real production constraints.
- Deployed MLflow-based LLM evaluation pipelines validating grounding accuracy under **500+ QPS** real-world query distributions, enabling low-latency production inference, load balancing and reliability for healthcare analytics systems.
- Reduced hallucinations using ChromaDB retrieval grounding, increasing answer faithfulness **32%** across healthcare and bioinformatics workflows, improving downstream decision reliability under sustained production usage conditions.

AI/ML Inference Engineer – LLM Serving & Optimization, FalconHiTech LLC | May 2025 – Dec 2025

- Designed and optimized vLLM inference pipelines, improving throughput **45%** and sustaining high-concurrency traffic while maintaining low-latency responses across production AI applications under continuous real-time usage.
- Implemented Python and C++ asynchronous batching, caching, sustaining, and profiling TensorFlow system behavior across **1200 QPS** while reducing queue wait times under inference workloads during peak production traffic periods.
- Improved vLLM serving stability, F1-score, increasing response accuracy by **14%**, and eliminating timeout failures across **6** enterprise environments supporting mission-critical LLM applications under sustained concurrent workloads.

Machine Learning - MLOps Engineer, Production ML Pipelines, TESCO LLC | May 2022 – Aug 2024

- Developed and deployed Kubernetes-hosted machine learning models by enabling intelligent automation and supporting **3x** adoption growth across enterprise production workflows under strict operational reliability constraints.
- Built MLflow and Kubernetes pipelines automating XGBoost for model training, evaluation, and deployment, reducing manual release effort **42%** while improving reproducibility across multiple production machine learning systems.
- Enhanced model performance, consistency, using production monitoring, reducing failure incidents **37%** and increasing stakeholder trust across **7** open source business workflows relying on continuously deployed machine learning systems.

PROJECTS

RAG System for Grounded Document QA (Python, FastAPI, GCP, Vector DB)

- Built a retrieval-augmented GenAI system to reduce hallucinations in knowledge-intensive queries within structured application workflows supporting complex information retrieval tasks across internal Transformers analytical use cases.
- Designed data ingestion, chunking, embedding, and vector retrieval pipelines integrated with an LLM backend, incorporating retrieval confidence thresholds and self-correction retries under production-scale operational constraints.
- Improved answer faithfulness and retrieval relevance by **~36%** through citation-backed RAG and retrieval tuning compared to vanilla LLM responses across multiple data sources and practical application scenarios in deployment.

ML / LLM Inference & Serving Optimization (Python, C++, PyTorch, Terraform)

- Optimized LLM inference pipelines to reduce latency and memory usage through dynamic batching, caching, and compute-cost tradeoff analysis under sustained high-throughput production workloads at scale environments globally.
- Implemented a high-throughput model serving on operating systems with asynchronous request handling, timeout protection, and performance profiling across QPS and p99 latency metrics under production traffic conditions at scale.
- Reduced end-to-end inference latency from **500ms to 350ms** and memory footprint by **~30–40%** through batching, caching, and serving optimizations under sustained production workloads at scale globally in worldwide environments.

LLM Evaluation & Hallucination Detection Framework (Python, LLMs, RAG, Evaluation Pipelines)

- Built an automated evaluation framework to assess hallucination rates and grounding quality in LLM and RAG-based systems under real-world query distributions across diverse production environments and deployment scenarios.
- Designed evaluation pipelines using answer–context alignment checks, retrieval confidence thresholds, and adversarial prompt variants across production-scale LLM systems metrics under diverse real-world operational prioritization.
- Reduced hallucinated responses by **~38%** and enabled systematic reliability benchmarking of LLM Evaluation output.

SKILLS

- **Languages:** Python, C++, Rust, Go, SQL; strong experience in model training, evaluation, and production AI systems.
- **AI & LLM Engineering:** Large Language Models (LLMs), A/B testing, Transformers, Natural Language Processing, RAG, Prompt Engineering, Supervised Fine-Tuning (LoRA/PEFT), LLM Evaluation & Hallucination Mitigation, R&.
- **Training & Optimization:** Distributed Training, Model Optimization, Quantization and Distillation, Optimization.
- **Frameworks & Libraries:** PyTorch, TensorFlow, Hugging Face, LangChain, TypeScript, FastAPI, CUDA, NLP, API.
- **LLM Systems & Serving:** Model Serving, Deep Learning, Dynamic Batching, Spark, Hadoop, Databricks, vLLM.
- **Data, Platform & MLOps:** Data Pipelines, Feature Engineering, Inference Optimization, Low-Latency Systems Experiment Tracking (MLflow), Model Monitoring & Observability, CI/CD, Docker, Kubernetes, AWS, GCP, Azure.
- **Databases & Integration:** Vector Databases, REST APIs, GPU, hybrid cloud, DevOps, Backend Services, GitHub.

EDUCATION

Master of Science in Computer Science, Northeastern University

Bachelor of Engineering in Computer Science, Osmania University

ACADEMIC & LEADERSHIP EXPERIENCE

Teaching Assistant, Cloud Computing | Northeastern University

- Supporting graduate-level cloud labs and system design sessions with faculty and students on cloud-based workflows.

Research Team Lead, Synthetic AI for AI Detection

- Owned the technical design of an ongoing Synthetic AI-based research project for multimodal AI-generated content detection, model development, evaluation, hallucination analysis, and problem-solving, achieving **98%** accuracy.

Chief Organizer, Google Developer Groups On Campus | MJCET

- Led the planning and execution of events and cloud workshops, decision-making, coordinating cross-functional student teams, and communication skills, driving a **3x** growth in student participation across multiple campus initiatives.